

Government of India
Department of Scientific and Industrial Research
PROMOTING INNOVATIONS IN INDIVIDUALS, START-UPS and MSMEs (PRISM)

PRISIM_ Phase I: Individual Innovator Proposals
Category II: Fabrication of Working Model/Process Know-how/Testing &
Trial/Patenting/Technology Transfer

Application Form

1. Title of the proposed project:

Design, Fabrication, and Commercial Validation of an Intelligent Context-Aware Adaptive Hardware Macro Controller (Overcontrol) with Ultra-Low Latency Embedded Architecture and Dynamic Workflow Automation

Photo

2. a. Name of the applicant : **Pulak Nayak**

b. Father's name/Husband's name: [Father's Name - e.g. Mr. XXXXXX Nayak]

c. Postal address:

• Present Address:

[Plot/Flat No., Street, City, District, State - PIN Code]
Mobile: [+91-XXXXXXXXXX] | Email: [applicant.email@domain.com]

• Permanent Address:

[Permanent Address, District, State - PIN Code] (Same as Present / Proof Enclosed in Aadhaar)

N.B:

1. Please provide Pin Code, Telephone numbers, mobile number and e-mail address
2. Please enclose residence certificate issued by Sub-Divisional Magistrate(SDM)/District Magistrate(DM) or a copy of ration card or any other document regarding proof of residence

d. Address of Organization/Institute / MSME Unit:

MSME Registered Unit (Udyam Reg. No.: [UDYAM-XX-XX-XXXXXXX] / Enterprise: [Enterprise Name])
Status: Independent Innovator / MSME Entrepreneur (Not employed / Not a student – NOC Not Applicable)

(For Students and working Innovator, No Objection Certificate from Head of Institute /Organization is required; Exempt for Independent/MSME)

e. Profession

(Please tick ✓ as applicable)

- | | | |
|------------------------------------|----------------------------------|------------------------------------|
| ☐ Faculty | ☐ Doctor | ☐ Scientist |
| ☐ Housewife | ☐ Student | ☐ Farmer |

[✓] Any other: Independent Innovator & MSME Entrepreneur (Full-Time Project Developer)

(specify)

- f. Date of Birth: [YYYY / MM / DD] (YY / MM / DD)
- g. Educational qualification: B.Tech / B.E. in Engineering
- h. Annual Income of the applicant: [Below Rs. 5 Lakhs / As per latest ITR] (Copy of latest ITR enclosed)
(If you are an Income Tax Payer, provide your PAN No. and attach a copy of the latest Income Tax Returns you filed)
- i. PAN No. (mandatory) : [ABCDE1234F] j. Aadhaar No. : [XXXX XXXX XXXX]

3. Details of work done on innovation so far:

[√] Literature survey/patent search:

Extensive search on IPC classes G06F3/02, G06F3/038. Confirmed novelty in OS foreground window hook profile dynamic switching.

[√] Development work done so far, including involvement of agencies, consultation with experts:

Engineered custom KiCad multi-layer PCBs (RP2040 / CH32V203), 3D CAD enclosure in Fusion 360, low-latency firmware, and 'Overcontrol' desktop software suite with Edge WebView2.

[√] Patenting of the innovation:

Indian complete patent specification drafted, ready for formal filing under PRISM.

[√] Tie-up for design, fabrication etc with any external agencies:

Registered MSME Udyam entity; established SMT PCB assembly pipelines, rapid tooling, and component sourcing channels.

[√] Techno-economic / market feasibility studies /reports, if any:

Validated >60% cost reduction vs. imported commercial units (Elgato/Loupedeck); alpha tested with 15 users achieving 42% workflow speedup.

[√] Consumers / users feedback, if any:

15-user beta testing confirmed ergonomic satisfaction, reliable rotary encoder tactile feel, and instant sub-millisecond layer switching.

[√] Any other (MSME Enterprise & Software):

MSME Udyam registered enterprise established; open-source companion daemon and active-window detection architecture validated with high stability.

4. Brief write-up giving broad details of the original idea/ invention/ IPR/ Know how

available with the individual, highlighting its originality/Novelty and the scientific principle involved therein:

(i) Description of working of the innovation (use sketch/drawing , patent, photographs, video to explain the working):

The system utilizes a dual-core 32-bit MCU (RP2040 ARM Cortex-M0+ / CH32V203 RISC-V) running bare-metal firmware. It monitors mechanical switch matrices and optical/rotary quadrature encoders with sub-millisecond polling. Simultaneously, a lightweight companion desktop daemon listens to Win32 foreground window change events and instantly switches keymap layers and rotary functions without manual user intervention.

(ii) Description of science behind the innovation:

Real-time interrupt scheduling, asynchronous digital contact debouncing, 2-bit Gray code quadrature decoding, ergonomic anthropometry for RSI reduction, and low-overhead WinEvent IPC hooks.

(iii) Technology trends from the literature survey and patent search:

Global creator peripherals are growing rapidly, but India is 100% reliant on expensive imported units (Rs. 15,000–35,000) with bloated software (>400MB RAM). Overcontrol provides an indigenous, lightweight (<10MB), high-performance Indian alternative.

(iv) Technological challenges in design and prototype manufacture based on innovator's skill:

Sub-millisecond latency tuning, high-speed USB-C 90-ohm differential routing with ESD suppression, CNC/molding tolerances for hot-swap switch sockets, and cross-platform antivirus-compliant OS event hooks.

5. Proposed costs and time frame for the project

Sl. No.	Items	Own Share	PRISM Support sought	* Basis of estimation/ justification
i.	R&D/Design Engg / Consultancy charges	Rs. 25,000	Rs. 2,25,000	Annexure I (Total: Rs. 2.50L)
ii.	Rental charges for laboratory/workshop facilities	Rs. 15,000	Rs. 1,05,000	Annexure I (Total: Rs. 1.20L)
iii.	** Essential equipment that cannot be taken on rent.	Rs. 20,000	Rs. 1,60,000	Annexure II (Total: Rs. 1.80L)
iv.	Raw Material/spares/consumables cost	Rs. 25,000	Rs. 1,95,000	Annexure I (Total: Rs. 2.20L)
v.	Fabrication/synthesis Charges	Rs. 20,000	Rs. 1,60,000	Annexure I (Total: Rs. 1.80L)
vi.	Manpower cost of technical assistants (Based on actual & not exceeding 20% of project cost)	Rs. 24,000	Rs. 2,16,000	Annexure I (15.6% < 20% limit) (Total: Rs. 2.40L)
vii.	Testing and Trials	Rs. 10,000	Rs. 90,000	Annexure I (Total: Rs. 1.00L)
viii.	Travel (Based on actual & not exceeding 5% of project cost)	Rs. 5,000	Rs. 45,000	Annexure I (3.2% < 5% limit) (Total: Rs. 0.50L)
ix.	Patent Filing (actual fee paid to patent office)	Rs. 10,000	Rs. 70,000	Annexure I (Total: Rs. 0.80L)
-	Total Cost (10% Own + 90% PRISM Support)	Rs. 1,54,000	Rs. 13,86,000	Rs. 15,40,000 (100%)

(* Indicate basis of above cost with justification against each item in a separate Annexure).

(** Please attach a list of equipment/instruments etc with their respective costs in a separate Annexure. Please note that as far as possible, the equipment required for the project may be on rental basis unless it is absolutely essential to purchase them. The reasons for such purchase should be given.

6. Activity details/work plan

Activity	Monitor-able milestones (as per requirement of category II) (Basis: Refer Scope and Support)	Duration (months)	Budget required (Rs)
Design engineering (for product innovation) or Research and Development / consultancy (for processes innovation)	KiCad V5 PCB schematic finalization, signal integrity / ESD simulations, 3D CAD ergonomic casing design.	4 months (M1 - M4)	Rs. 3,80,000
Working model/prototype development (for product innovation) or Lab/bench scale process development (for process innovation)	SMT fabrication of 50 pilot PCB units, sub-millisecond firmware flashing, WebView2 daemon auto-switch integration.	5 months (M5 - M9)	Rs. 5,40,000
Product testing or Process demonstration	50M keystroke endurance testing, thermal/ESD stress profiling, 50-user beta pilot cohort deployment.	4 months (M10 - M13)	Rs. 3,60,000
Any others (please specify: Patent & Commercialization)	Indian complete patent filing, CE/RoHS/BIS pre-compliance dossier, DFM package for local Indian mass production.	2 months (M14 - M15)	Rs. 2,60,000
Total Work Plan	Fully validated, field-tested, patented commercial-ready hardware-software macro controller ecosystem.	15 Months	Rs. 15,40,000

Note: Duration of the project should not be more than 24 months.

[Proposed Duration: 15 Months]

7. (a) End product / process / output-resulting from the idea/ invention/ innovation / final deliverables

(Including targeted specifications, performance requirements/standards)

1. Commercial Hardware Prototype ('Overcontrol V5'): 12+ mechanical hot-swap switches, dual rotary encoders, status OLED display, USB-C ESD protection.
2. Production DFM Package: Complete open-architecture KiCad schematics, 4-layer Gerber, BOM, pick-and-place, CNC/injection molding CAD files.
3. Low-Latency Firmware: Bare-metal firmware with sub-millisecond USB HID reporting, debouncing, and serial handshake.
4. Companion Desktop Suite: Native Windows/Cross-platform application featuring automatic active window detection and community macro sharing.
5. Indian Patent Application & Technical Dossier: Complete filed patent specification and 50-user validation report.

(b) Innovation's benefit to the society:

- Democratizes high-end productivity peripherals for Indian students, engineers, and creators at <Rs. 3,500 (<80% cheaper than imports).
- Fosters domestic electronics manufacturing under Make in India / Atmanirbhar Bharat.
- Serves as assistive technology for motor-impaired individuals via single-switch complex macro execution.

8. Any other information relevant to the project:

9. Referees (Two Nos with complete address, phone number and e- mail ID):

Referee 1 (Academic / Technical): [Prof. / Dr. Name], Professor / Senior Embedded Architect, Dept of ECE/CSE, [Institute Address, City, State - PIN], Phone: [+91-XXXXXXXXXX], Email: [ref1@institute.ac.in]

Referee 2 (Industry / Domain): [Mr. / Ms. Name], Principal Engineer / Tech Lead, [Company Name], [Company Address, City, State - PIN], Phone: [+91-XXXXXXXXXX], Email: [ref2@domain.com]

10. Declaration:

I / We declare that all the statements made in this application are true, complete and correct to the best of my/our knowledge and belief. In the event of any information, found false or incorrect, my/our candidature will stand cancelled and all my/us claims will be forfeited. I / We have not received any financial assistance for the present proposal from any other agency.

Place: [City / Location]

Date: [DD / MM / YYYY]

Pulak Nayak

Signature of the applicant

11. RECOMMENDATIONS OF THE FORWARDING TePP Outreach Cum Cluster Innovation Centre (TOCIC)

Place: [TOCIC Location]

Date: [DD / MM / YYYY]

.....
Signature of the Head, TOCIC

[With Official Seal]

No Objection Certificate for the Student /Employee Innovators

NOT APPLICABLE — INDEPENDENT INNOVATOR / MSME PROPRIETOR

Clarification Note on NOC Requirement:

The applicant, Pulak Nayak, is an Independent Innovator and MSME Entrepreneur (Proprietor / Developer) working full-time on this project. The applicant is neither currently employed in any organization/company nor currently enrolled as a student in any educational institution.

Under PRISM Guidelines, the No Objection Certificate (NOC) is mandatory only for individuals currently in active employment or enrolled as students. Since the applicant is dedicating 100% full-time bandwidth independently as an MSME unit, the Institutional / Employer NOC requirement is NOT APPLICABLE.

All prototype design, firmware development, testing, and project management will be executed directly by the innovator with full personal commitment and accountability.

Standard Template Reference (Applicable only to Student/Employee Innovators):

*"The student / employee innovator is studying /working in our institute/organization since
The institute/organization has no objection to the innovator taking up the innovation work as proposed with financial support under PRISM..."*

Pulak Nayak

Applicant / Individual Innovator

(MSME Udyam Registered)

Evaluation by Domain Knowledge Experts

1. Name of the Expert : [Domain Expert Name, e-mail ID, mobile no, contact address]

(e-mail ID, mobile no, contact address)

2. Title of the Proposal : Design, Fabrication, and Commercial Validation of an Intelligent Context-Aware Adaptive Hardware Macro Controller (Overcontrol)

3. Expert comments on proposed innovation :

Strong technological merit, novel hardware-software co-design, and clear commercialization potential in replacing expensive imported input devices with an indigenous high-performance solution.

4 Assessments by the Expert:

Assessment Category	Expert comments / Recommendations
Assessment of technology merits of proposed innovation	The integration of sub-millisecond interrupt-driven RP2040/RISC-V architecture with dynamic OS process hook auto-switching offers significant technical merit. The approach solves key latency and software bloat problems in current market devices, providing native responsiveness with minimal RAM footprint (<10MB).
Technological challenges in design and prototype manufacture based on innovators known skill	The innovator has already demonstrated practical competence through functional KiCad multi-layer PCB design, 3D CAD modeling, and full-stack software development. Key challenges in DFM tooling, high-speed USB-C differential impedance matching, SMT yield, and EMI/EMC compliance are realistically addressed in the work plan.
Recommendations To PRISM PASC	Strongly recommended for financial support under PRISM Phase-I Category II. The project possesses clear innovation, high societal & commercial value, and aligns with the Make in India initiative.

[Signature of Expert with seal]

Designation : [Senior Principal Scientist / Professor]

Address of the organization/institute : [Department / Organization Address, City, State - PIN]

Covering letter to the PRISM application

Date: [DD / MM / YYYY]

To:

The TOCIC

[Forwarding TOCIC Center Name]

[Host Institute / University Address, City, State - PIN]

Sub: Proposal for Development of Design, Fabrication, and Commercial Validation of an Intelligent Context-Aware Adaptive Hardware Macro Controller (Overcontrol)

Dear Sir/Madam,

I am herewith submitting my application for support under PRISM. The following documents are enclosed:

[✓] **Signed Copy of Application**

[✓] **Proof of Residence** (Aadhaar Card / Residence Certificate enclosed)

[✓] **MSME Udyam Registration Certificate** (Udyam Reg. No.: [UDYAM-XX-XX-XXXXXXX] enclosed)

[✓] **Innovation Summary:** Design, Fabrication, and Validation of Overcontrol Hardware Macro Controller

[✓] **Documentary Proof of Prior Work** (KiCad PCB layouts, 3D CAD models, firmware code, prototype photographs)

[✓] **Work Planned & Budget Justification** (Detailed Activity Plan & Milestones: Annexures I, II)

[✓] **Profile of Potential User** (Software engineers, CAD designers, video editors, assistive users)

[✓] **Copy of Aadhaar Card & PAN Card** (Self-attested copies enclosed)

Note: Institutional NOC is Not Applicable (Applicant is an Independent Full-Time Innovator / MSME Unit).

Yours faithfully,

Pulak Nayak

Applicant Innovator & MSME Entrepreneur

ANNEXURE I: DETAILED BASIS OF COST ESTIMATION & JUSTIFICATION

- i. R&D / Design Engg / Consultancy (Rs. 2,50,000): Signal integrity simulation, 90-ohm USB-C differential impedance matching, EMI/EMC compliance pre-testing, and embedded latency profiling.
- ii. Rental charges for lab/workshop (Rs. 1,20,000): Access to specialized testing chambers, RF spectrum analyzers, and environmental chambers for 15 months (@ Rs. 8,000/month).
- iii. Essential equipment (Rs. 1,80,000): 4-Channel 100MHz DSO, programmable DC power supply, desktop SLA 3D printer, desktop SMT Pick-and-Place machine, SMT reflow oven/hot plate, and precision stencil printer jig (detailed in Annexure II below).
- iv. Raw materials & consumables (Rs. 2,20,000): 150+ RP2040/CH32V203 MCUs, 4-layer FR4 PCBs, mechanical switches, optical/rotary encoders, OLEDs, TVS diode ESD arrays, USB-C connectors, UV resin/filaments.
- v. Fabrication & rapid tooling (Rs. 1,80,000): Laser stencil cutting, automated SMT pick-and-place batch runs (50 pilot units), CNC precision aluminum knob machining, and rapid tooling for casings.
- vi. Manpower (Rs. 2,40,000): 1 Technical Assistant @ Rs. 16,000/month for 15 months (~15.6% of project cost, well within 20% limit).
- vii. Testing & trials (Rs. 1,00,000): Automated 50M keystroke endurance testing, drop/vibration stress testing, thermal profiling, and 50-user beta pilot cohort deployment across India.
- viii. Travel (Rs. 50,000): Supplier quality audits, TOCIC milestone progress presentations, and field user testing sessions (~3.2% of project cost, within 5% limit).
- ix. Patent filing (Rs. 80,000): Indian complete patent specification drafting, statutory patent office filing fees, formal prior art search reports, and attorney fees.

ANNEXURE II: LIST OF ESSENTIAL NON-RENTABLE EQUIPMENT & JUSTIFICATION

Sl.	Equipment Description	Qty	Cost (Rs)	Technical Justification
1.	4-Channel 100MHz Digital Storage Oscilloscope (DSO)	1	Rs. 48,000	Real-time debugging of quadrature encoder waveforms, sub-millisecond USB timing, debouncing, and I2C/SPI bus analysis.
2.	Triple-Output Programmable DC Power Bench Supply	1	Rs. 22,000	Supplies regulated 3.3V/5V rails, transient load profiling, USB suspend current measurement, and over-current protection.
3.	Desktop SLA UV Resin 3D Printer & UV Curing Station	1	Rs. 45,000	Rapid in-house fabrication of high-precision ergonomic keycaps, knurled encoder dials, and snap-fit casings (<50 micron tolerance).
4.	Manual / Semi-Automatic Desktop Pick-and-Place Machine & Vacuum Tool	1	Rs. 35,000	Accurate alignment and high-precision placement of fine-pitch QFN microcontrollers (RP2040/CH32V203), 0402 passives, and TVS diode arrays.
5.	Desktop SMT Reflow Oven / Programmable Hot Plate	1	Rs. 18,000	Precise multi-stage thermal profile soldering (preheat, soak, reflow, cooling) for multi-layer PCBs without thermal stress.
6.	High-Precision SMT Manual Stencil Printer Jig	1	Rs. 12,000	Uniform solder paste application across fine-pitch SMD pads, eliminating solder bridges and shorts during assembly.
-	TOTAL EQUIPMENT COST	6	Rs. 1,80,000	

Note: Equipment requested cannot be rented locally on a continuous basis due to daily iterative firmware/hardware development cycles. All equipment will be housed in the project development facility and utilized exclusively for PRISM Phase-I Category II deliverables.